

Part A: Negative Number Operations*Evaluate each expression completely.*

1. $8^3 - (-13) \cdot 10$

6. $(-6)^2 - (-5) \cdot 9$

2. $-16 - 3 \cdot (2 + (-16))$

7. $(-5)^3 - 8 \cdot 12$

3. $-9 - 7 \cdot (2 + (-9))$

8. $8^3 - 5 \cdot 2$

4. $(-2)^3 - (-3) \cdot 8$

9. $14 - 5 \cdot (2 + 14)$

5. $(6 + (-13))^2 \div 1$

10. $(3 + 4)^2 \div 1$

Part B: Negative Number Operations*Evaluate each expression completely.*

1. $-3 - 33$

6. $10 \cdot (-11)$

2. $13 + (-32) - (-14)$

7. $11 \cdot (-9)$

3. $11 - 29$

8. $6 + 10 \cdot 12$

4. $5 + 12 + (-9)$

9. $30 \div 5 - 16$

5. $45 \div (-15)$

10. $-60 \div 6 - (-9)$

Part C: Solving Linear Equations*Solve each equation for x . Show your steps.*

1. $7x = -63$

2. $2x = -4$

3. $2x + 3 = -13$

7. $8x + 12 = -7x + \frac{183}{4}$

4. $7x + 6 = 41$

8. $9x - 1 = -6x + 179$

5. $4(9x + 3) - 11 = 397$

9. $\frac{x - 6}{2} - 10 = -\frac{39}{2}$

6. $5(7x + 6) - 11 = 54$

10. $\frac{3x - 24}{4} = \frac{2x + 5}{12}$

Part D: Linear Inequalities*Solve each inequality for x .*

1. $x - 6 \leq 4$

6. $9x + 1 < 2x + 85$

2. $x + 9 > -4$

7. $-x - 6 \geq -1$

3. $-9x + 2 \geq 20$

8. $\frac{x + 12}{2} - 4 < -4$

4. $-7x + 2 > -19$

9. $\frac{3}{2}x - 5 < \frac{5}{4}x - \frac{19}{4}$

5. $-11x + 2 < -5x + 32$

10. $-\frac{1}{4}x - 3 < -\frac{1}{2}x - \frac{27}{4}$

Part E: Exponent Rules*Simplify. Write answers with positive exponents.*

1. $3x^6 \cdot 2x^4$

6. $(8x^8)^3$

2. $2x^3 \cdot x^4$

7. $6 \cdot (10^1)^{-1}$

3. $\frac{2x^8}{x^5}$

8. $\frac{6}{3^{-3}}$

4. $\frac{2x^{10}}{x^3}$

9. $5^4 \cdot 5^{-4}$

5. $(3x^6)^2$

10. $\frac{5x^8y^8}{x^7y^6}$

Part F: Simplifying Expressions*Simplify each expression.*

1. $-7x + 6 + x + 8$

4. $9(9x - 9) + 9x - 12$

2. $5x + 5 + 3x - 7$

5. $8(4x + 8) + 12x + 7$

3. $6x + 4 - x + 9$

6. $5(8x + 4) + 3x + 4$

Part G: Adding and Subtracting Linear Expressions*Add or subtract, then simplify.*

1. $(-12x + 7) + (4x + 9)$

2. $(-11x + 10) - (-3x - 12)$

Part H: Factoring Linear Expressions*Factor out the greatest common factor.*

1. $28x - 35$

2. $-30x - 42$

Answer Key

Part A: Negative Number Operations

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|--------|---------|--------|
| 1. 642 | 5. 49 | 9. -66 |
| 2. 26 | 6. 81 | 10. 49 |
| 3. 40 | 7. -221 | |
| 4. 16 | 8. 502 | |

Part B: Negative Number Operations

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|--------|---------|--------|
| 1. -36 | 5. -3 | 9. -10 |
| 2. -5 | 6. -110 | 10. -1 |
| 3. -18 | 7. -99 | |
| 4. 8 | 8. 126 | |

Part C: Solving Linear Equations

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|-------------|----------------------|--------------|
| 1. $x = -9$ | 5. $x = 11$ | 9. $x = -13$ |
| 2. $x = -2$ | 6. $x = 1$ | 10. $x = 11$ |
| 3. $x = -8$ | 7. $x = \frac{9}{4}$ | |
| 4. $x = 5$ | 8. $x = 12$ | |

Part D: Linear Inequalities

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|----------------|----------------|---------------|
| 1. $x \leq 10$ | 5. $x > -5$ | 9. $x < 1$ |
| 2. $x > -13$ | 6. $x < 12$ | 10. $x < -15$ |
| 3. $x \leq -2$ | 7. $x \leq -5$ | |
| 4. $x < 3$ | 8. $x < -12$ | |

Part E: Exponent Rules

- | | | |
|--------------|------------------|-------------|
| 1. $6x^{10}$ | 5. $9x^{12}$ | 9. 1 |
| 2. $2x^7$ | 6. $512x^{24}$ | 10. $5xy^2$ |
| 3. $2x^3$ | 7. $\frac{3}{5}$ | |
| 4. $2x^7$ | 8. 162 | |

Part F: Simplifying Expressions

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|--------------|---------------|---------------|
| 1. $14 - 6x$ | 3. $5x + 13$ | 5. $44x + 71$ |
| 2. $8x - 2$ | 4. $90x - 93$ | 6. $43x + 24$ |

Part G: Adding and Subtracting Linear Expressions

- | | |
|--------------|--------------|
| 1. $16 - 8x$ | 2. $22 - 8x$ |
|--------------|--------------|

Part H: Factoring Linear Expressions

1. $7(4x - 5)$

2. $6(-5x - 7)$